The Anatomy of Firm Scope*

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Abstract

I show that firm scope operates along two economically distinct dimensions. Text-based product-market clusters measure related scope—closely linked sub-markets sharing a technological substrate—while the Compustat segment count measures distant scope—operationally separable lines of business. Using a panel of U.S. public firms over 1990–2019, I estimate local projections on patent-value innovation shocks and find that these dimensions govern opposite regimes. Broad related scope generates synergy: firms tilt intangible investment toward R&D, attract rival entry, and convert innovation into persistent productivity and markup gains. High distant scope generates friction: rents dissipate, productivity gains vanish, and intangible accumulation shifts toward brand and organisational capital. The superstar-firm profile emerges from the structure, not the level, of scope.

Keywords: firm scope, diversification, intangibles, productivity, markups

JEL Classification: L25, O31, E22.

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1. Introduction

Modern firms rarely confine themselves to a single product market, yet the *structure* of their scope varies sharply. Some firms operate across many closely related sub-markets that share technological foundations, distribution channels, and customer bases. Others span operationally distant business domains that share little common substrate. The distinction between related and unrelated diversification has been a central concern of corporate research since Panzar and Willig (1981) and Teece (1980), and it continues to shape work on innovation synergies (Bena and Li, 2014) and product-market relatedness (Hoberg and Phillips, 2010; Hoberg and Phillips, 2016). This paper asks how the structure of firm scope—whether a firm's markets are closely related or operationally distant—shapes its productivity, markups, intangible investment, and exposure to competitive threat.

To answer this question, I draw on two complementary measures of scope. The text-based dataset of Hoberg and Phillips (2025) resolves a firm's activity into fine-grained product-market clusters that typically lie close to one another in product space; it measures *related scope*, the closely related sub-markets a firm operates across. The Compustat Historical Segment file aggregates the same firm into the broad reporting units that managers disclose under SFAS 131. These segments are constructed precisely to partition activity into operationally distinct lines of business, so the segment count measures *distant scope*, the number of largely non-overlapping domains a firm spans. For instance, Hoberg–Phillips resolves Tesla into eleven product-market clusters—covering automotive subsystems (brakes, trim, axles, chassis), automotive safety, car dealerships, batteries, solar and renewable energy, utility-scale electric power, smart metering, power electronics, hardware and software solutions, and ticketing systems—all anchored in a common electrochemical and energy-systems substrate. Compustat aggregates the same firm into two operationally distant segments: Automotive and Energy Generation & Storage.[1] The two datasets are not redundant alternatives; they are complementary windows on different

[1]See Table A1 for details.

layers of scope.

I bring these measures to firm-level Compustat data spanning fiscal years 1990 to 2019. Distant scope is measured by the number of business segments a firm reports in the Compustat Historical Segment file, related scope by the text-based product-market clusters of Hoberg and Phillips (2025), and competitive threat by the product-market fluidity index of Hoberg *et al.* (2014), which captures how quickly rivals are encroaching on a firm's product space. To these I add firm-level measures of total factor productivity, markups, and the composition of intangible capital, separating research-based intangibles (R&D) from embedded intangibles such as brand and organisational capital. The empirical design follows the local-projection method of Jordà (2005), applied to the patent-value innovation shocks of Kogan *et al.* (2017), which measure the economic value of a firm's patents from the stock-market reaction to their grant. This traces the dynamic response of each outcome to a well-identified innovation shock and, crucially, allows the response to differ along the two dimensions of scope.

The two layers of scope rest on different economics. Related scope rewards transferable intangibles such as R&D, whose returns compound across closely related submarkets; distant scope rewards embedded intangibles such as brand and organisational capital, which attach to the firm rather than to a technology and so bridge domains that technical knowledge cannot. Related scope is therefore a regime of synergy, distant scope a regime of friction, and the paper's central question is which regime governs the appropriation of innovation rents.

The synergy and friction predictions are borne out. Firms with broad related scope absorb innovation aggressively: they accelerate investment in transferable intangibles, attract rival entry, and convert the shock into persistent productivity and markup gains. Firms spread across many operationally distant segments respond weakly on every margin, and their productivity response is indistinguishable from zero. Innovation rents are appropriated where related scope supplies synergistic deployment outlets, and they dissipate where activity is fragmented across operationally disjoint domains in which R&D does not transfer.

The combined message is that *scope is not a scalar*. Two firms with the same number of markets can inhabit opposite economic regimes—one of synergy, one of friction—depending on whether those markets are closely related sub-markets or operationally distant segments. Firms that combine broad related scope with few distant segments exhibit the empirical profile—high productivity, high and persistent markups—that Autor *et al.* (2020) and De Loecker *et al.* (2020) associate with superstar firms and the rise of market power. Those studies characterise such firms by their outcomes rather than by their scope; the structure of scope offers an account of what underlies the profile. A shared technological substrate lets R&D compound across synergistic sub-markets, raising productivity, sustaining markups, and concentrating intangible accumulation in the input whose returns the firm can appropriate across its full product-market portfolio. Firms that combine many distant segments with narrow related scope display the configuration of the diversified conglomerates studied by Lang and Stulz (1994), Berger and Ofek (1995), and Schoar (2002): the absence of a shared substrate dissipates the appropriability of R&D and leaves brand and organisational capital as the only intangibles able to bridge operationally disjoint domains, compressing productivity and markups. The cross-firm gradients in productivity, market power, and intangible composition are therefore joint consequences of the underlying structure of scope rather than independent margins of firm behaviour.

Related literature. The paper contributes to four bodies of work. The first is the classical literature on economies of scope and corporate diversification (Panzar and Willig, 1981; Teece, 1980; Helfat and Raubitschek, 2000; Lang and Stulz, 1994; Berger and Ofek, 1995; Schoar, 2002; Maksimovic and Phillips, 2002; Villalonga, 2004; Custódio, 2014). The economies-of-scope tradition emphasises that closely related sub-markets share technological inputs—R&D, know-how, and intermediate goods scale across linked sub-markets without proportional duplication of fixed costs—whereas operationally distant domains share few such inputs. I show that the diversification discount is a feature of distant scope, that related scope generates the opposite pattern, and that the two effects coexist within the same firm and the same sample once the dimensions are separated. The second is the literature on rising markups and superstar firms (De Loecker *et al.*, 2020; Autor *et al.*, 2020; Crouzet and Eberly, 2019; Crouzet and Eberly, 2023). The joint movement of markups and productivity at the core of recent concentration narratives is driven by firms with broad related scope and few distant segments, a configuration that fits an increasing-dominance reading of contemporary industry structure. The third is the literature on intangible capital (Corrado *et al.*, 2009; Eisfeldt and Papanikolaou, 2013; Eisfeldt and Papanikolaou, 2014; Peters and Taylor, 2017; Haskel and Westlake, 2017). What travels across operationally distant domains is not technical knowledge but firm-specific intangible capital—brand reputation, managerial practices, and organisational routines—whereas research-based intangibles compound only where sub-markets are technologically linked. The structure of firm scope thus emerges as a first-order determinant of intangible composition: firms tilt toward R&D when scope is related and toward brand and organisational capital when scope is distant, consistent with the differential transferability of these inputs across product markets. The fourth is the empirical innovation literature built on patent-value shocks (Kogan *et al.*, 2017; Bena and Li, 2014). The appropriability of innovation depends on the structure of scope rather than its level: related scope amplifies innovation rents, and distant scope dissipates them.

Paper organisation. The remainder of the paper proceeds as follows. Section 2 describes the data sources and presents summary statistics on the two scope measures. Section 2.1 details the firm-level measurement of productivity, markups, and the components of intangible capital. Section 3 estimates the dynamic responses to patent-value innovation shocks across the two scope dimensions. Section 4 concludes.

2. Data and Summary Statistics

Compustat Fundamentals provides comprehensive firm-level financial information for publicly listed companies in North America,[2] with longitudinal coverage of balance-sheet items, income-statement components, and cash-flow data. I use Compustat to construct the firm-level measures of markups, productivity, and intangible composition described in Section 2.1, and as the spine onto which the two scope datasets are merged. The sample covers fiscal years 1990–2019 and excludes firm-years with missing or non-positive R&D, SG&A, employment, or sales.

Measuring distant scope—the operationally separable lines of business a firm spans—requires data on the broad segments that firms report to investors as substantively separable. I obtain this from the Compustat Historical Segment file, which compiles firms' mandated segment-level disclosures from 1976 to the present.[3] Compustat assigns a unique segment identifier (`sid`) to each segment a firm reports, and I define distant scope as the number of distinct business segments reported in a given year. Under the management approach of SFAS 131, segment granularity is determined internally by the firm, and the standard requires that segments correspond to components of the enterprise whose operating results management reviews separately to assess performance and allocate resources. In practice, this yields broad partitions that distinguish operationally and informationally separable lines of business rather than fine-grained product variety, so the segment count captures diversification across domains rather than within-domain

[2] Compustat North America includes foreign firms cross-listed in the U.S. via American Depositary Receipts (ADRs), such as Toyota and Unilever. These firms are subject to SEC regulation, comply with U.S. reporting requirements, compete in U.S. product markets, and obtain patents from the U.S. Patent and Trademark Office, making them economically relevant for this study. Excluding them reduces the sample size by approximately 15% without altering the main conclusions, so they are retained throughout.

[3] U.S. public firms are required to disclose segment-level information in their 10-K filings. From 1976 to 1997, these disclosures were governed by SFAS No. 14, under which segments were defined using an *industry approach*. From fiscal year 1998 onward, segments are reported under SFAS No. 131. Firms without segment records in a given year are excluded from the sample.

breadth.[4]

Measuring related scope—the fine-grained product-market clusters a firm spans within its operational domain—requires data of a different kind. I use the Hoberg–Phillips text-based scope dataset (Hoberg and Phillips, 2025), which applies natural-language processing to the product descriptions firms file in their annual 10-K filings. The construction embeds each firm's product description in a high-dimensional vector-space representation of the U.S. product market and assigns the firm to a set of product-market clusters based on its proximity to other firms in that space. The variable `d2vscope` reports the number of distinct clusters a firm occupies in a given year, and I use it as the measure of related scope throughout. Because the clusters are recovered from product-description language rather than from managerial reporting, they are typically narrower than Compustat business segments and capture within-domain product variety that the segment count is constructed to abstract away from. Table A1 illustrates the contrast for Tesla, Inc., which Compustat aggregates into two distant segments while Hoberg–Phillips resolves into eleven related product-market clusters.

To capture the competitive pressure firms face, I use the product-market fluidity metric of Hoberg *et al.* (2014), which measures how rapidly rivals in similar product markets change their offerings. The metric is constructed using natural-language processing on product descriptions from firms' annual 10-K filings, tracking year-over-year changes in how companies describe their business activities. A high fluidity score indicates that rivals are rapidly reconfiguring their market positions, intensifying the competitive pressure the firm faces.

To trace how firm outcomes evolve around innovation events, I use the patent-value dataset of Kogan *et al.* (2017), which provides patent identifiers, filing and grant dates, firm identifiers, forward citations, and a market-based estimate of patent value. The patent-

[4]The Historical Segment dataset reports several segment classifications at the firm-year level, including business, operating, and geographic segments. Beginning in 2015, the WRDS Historical Segment files additionally provide product–service (PD_SRVC) information, which directly identifies the products and services associated with each segment. To construct a consistent measure of firm scope, I therefore use business segments prior to 2015 and PD_SRVC-based segment definitions thereafter.

value measure is constructed from stock-market reactions within a narrow event window around patent grant dates, capturing the surprise component of the market's response to each grant. The short-window design isolates the incremental information content of the grant itself, abstracting from anticipation effects and concurrent firm-level news, and provides the source of plausibly exogenous innovation variation used in the local-projection estimates of Section 3.

Table 1. Summary Statistics

Variable	N	Mean	SD	P25	Median	P75
Sale	71,139	3,398,985	17,129,640	22,203	121,055	846,755
Emp	71,139	9,814	37,290	135	597	3,702
AT	71,136	4,502,256	23,855,832	26,457	142,066	981,467
XRD	71,139	147,233	746,922	1,589	7,513	35,977
XSGA	71,139	641,985	2,651,924	11,640	44,070	208,185
Productivity (OP)	65,211	20.416	23.416	7.861	11.106	24.321
Markup (OP)	64,591	1.543	1.042	0.988	1.210	1.665
Productivity (ACF)	51,847	30.191	49.551	6.410	8.391	36.216
Markup (ACF)	45,510	1.275	0.768	0.885	1.080	1.417
Productivity (GNR)	65,843	11.888	22.762	0.880	1.562	7.911
Markup (GNR)	64,647	1.291	0.914	0.826	1.014	1.382
Competitive Threat	51,975	6.781	3.211	4.442	6.370	8.619

Notes: This table reports summary statistics for the three estimation samples and covers Compustat Fundamentals Annual restricted to firm-years with positive sales, employment, R&D, and SG&A.

Table 1 reports pooled summary statistics for the main variables used in the analysis over fiscal years 1990–2019. The underlying Compustat sample exceeds 70,000 firm-year observations on the raw size measures (sales, employment, total assets, R&D, and SG&A) and displays the pronounced right-skew familiar from the Compustat cross-section: median sales and total assets are roughly two orders of magnitude smaller than their means. Firm-level productivity and markup are reported under three production-function estimators—Olley and Pakes (1996)(OP), Ackerberg *et al.* (2015)(ACF), and Gandhi *et al.* (2020)(GNR). Their precise construction is detailed in Section 2.1. As is standard in this literature, the levels differ across estimators while the cross-sectional dispersion is broadly comparable in proportional terms; the empirical results that follow are reported under the Olley–Pakes baseline, with the alternative estimators in the appendix. Competitive threat, measured by the Hoberg *et al.* (2014) fluidity index, has a mean of 6.78 and standard deviation of 3.21, in line with the moments reported in prior work. Because the two scope datasets cover overlapping but non-identical subsets of Compustat, the merged samples are smaller; Appendix Table A4 reports summary statistics for the distant-scope and related-scope merged samples separately.

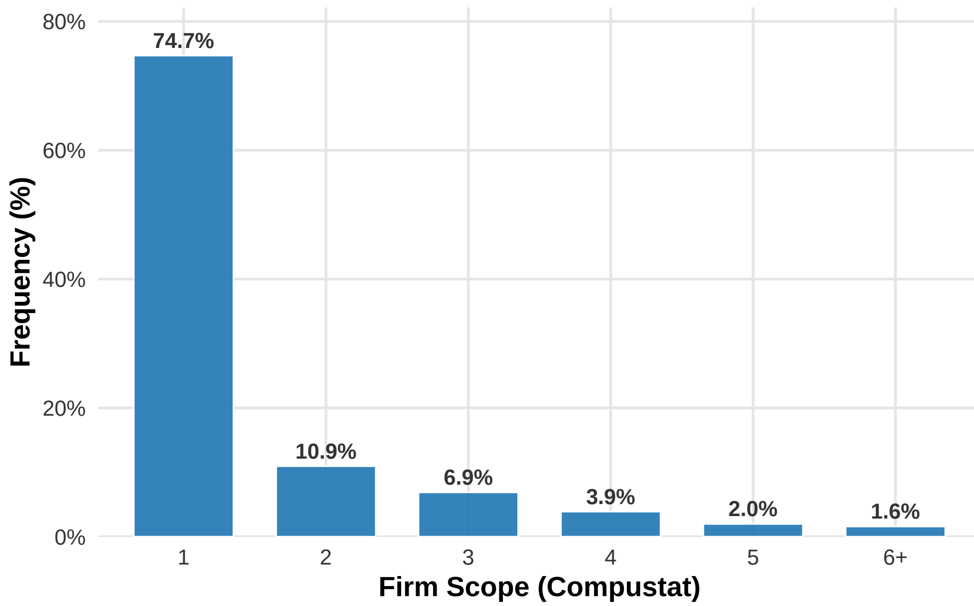

Figure 1. Firm Distribution across Scope (Compustat)

Note: The figure shows the distribution of firm-year observations across the number of reported business segments (*sid_count*). Observations with zero segments are excluded. Values greater than five are grouped into the "6+" category.

Figures 1 and 2 plot the pooled firm-year distributions of the two scope measures. The shapes differ starkly. Distant scope is heavily concentrated at a single segment: 74.7% of firm-year observations report one business segment, 10.9% report two, and the share declines sharply thereafter, with only 7.5% of observations reporting four or more segments. Operational diversification across separable lines of business is, in the panel, confined to a narrow upper tail. Related scope, in contrast, is far more dispersed. The mode at one cluster accounts for only 12.9% of firm-year observations; mass is distributed smoothly across the interior of the support, with roughly 59% of observations falling between two and ten clusters, and a non-trivial right tail in which 6.0% of observations occupy twenty or more clusters. The two measures therefore describe distinct margins of expansion: firms remain predominantly narrow across operationally distant domains while spreading broadly across related product-market clusters.

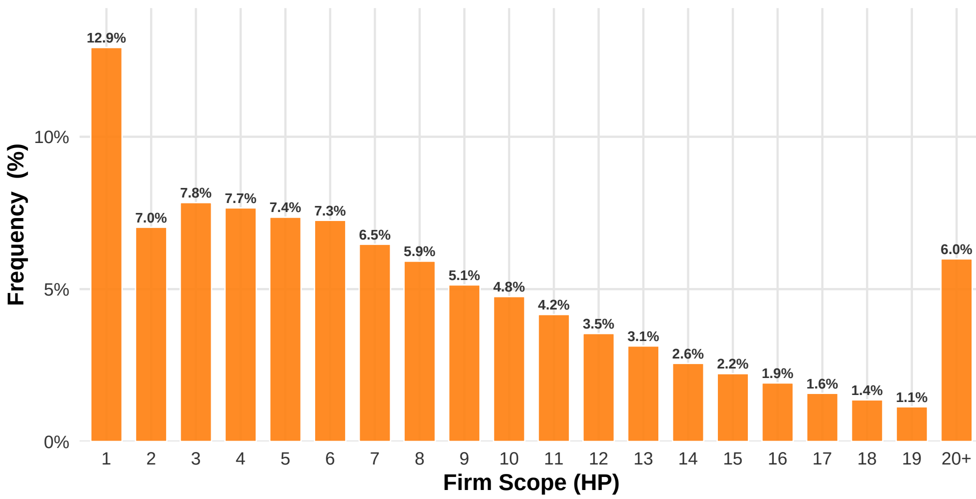

Figure 2. Firm Distribution across Scope (Hoberg-Phillips)

Note: The figure shows the distribution of firm-year observations across the firm scope measure (*d2vscope*), which captures the breadth of a firm's product market activity. Observations with a scope value of zero are excluded. Values greater than nineteen are grouped into the "20+" category.

2.1. Measurement

I estimate firm-level total factor productivity using the two-stage proxy-variable approach of Olley and Pakes (1996), which addresses the simultaneity between input choices and unobserved productivity by exploiting the monotonicity of investment in productivity conditional on capital. Consider the gross-output production function

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + \omega_{it} + \varepsilon_{it}, \tag{1}$$

where y_{it}, l_{it}, k_{it}, and m_{it} denote log sales, log employment, log capital stock, and log intermediate inputs for firm i in year t;[5] ω_{it} is the productivity shock observed by the firm when choosing inputs; and ε_{it} is an i.i.d. disturbance. OLS estimates of (1) are biased because flexible-input choices respond to ω_{it}. The proxy-variable procedure resolves this simultaneity by using observable investment, conditional on capital, to recover the unobserved ω_{it}.

First stage. Under the assumption that the capital-investment policy $i_{it} = f_t(\omega_{it}, k_{it})$ is strictly increasing in ω_{it} holding k_{it} fixed, the policy function is invertible and productivity can be written as $\omega_{it} = h_t(k_{it}, i_{it})$, where i_{it} denotes log capital expenditure (CAPX). Substituting this inversion into the production function delivers the first-stage estimating equation

$$y_{it} = \beta_l l_{it} + \beta_m m_{it} + \phi_t(k_{it}, i_{it}) + \varepsilon_{it}, \qquad \phi_t(k_{it}, i_{it}) \equiv \beta_k k_{it} + h_t(k_{it}, i_{it}), \tag{2}$$

where ϕ_t is approximated by a third-order polynomial in (k_{it}, i_{it}) interacted with year fixed effects. The first stage identifies β_l, β_m, and the composite function ϕ_t.

Second stage. The capital coefficient β_k is identified from the Markov structure of productivity. Under $\omega_{it} = g(\omega_{i,t-1}) + \xi_{it}$, the innovation ξ_{it} is orthogonal to capital chosen at $t-1$—because capital is decided one period in advance and so cannot respond to the current-period innovation—yielding the moment condition

$$\mathbb{E}[\,\xi_{it}(\beta_k) \cdot k_{it}\,] = 0, \tag{3}$$

where $\xi_{it}(\beta_k)$ is constructed as the residual from a non-parametric regression of $\hat{\omega}_{it}(\beta_k) \equiv \hat{\phi}_t - \beta_k k_{it}$ on its first lag. I correct for selection bias from firm exit by including the estimated survival probability in the second-stage moment, as in OP. Firm-level productivity

[5] The Compustat counterparts are SALE (sales), EMP (employment), PPENT (net property, plant, and equipment), and intermediate inputs constructed as COGS less the imputed wage bill, following standard practice. All variables are deflated using BEA gross-output deflators at the two-digit NAICS level.

is then recovered as $\hat{\omega}_{it} = \hat{\phi}_t - \hat{\beta}_k k_{it}$.[6]

Firm-level markups follow De Loecker *et al.* (2020): $\sigma_{it} \equiv \hat{\beta}^V_{it} / s^V_{it}$, where $\hat{\beta}^V_{it}$ is the output elasticity of the variable input (intermediates) recovered from the production-function estimation above, and s^V_{it} is the variable input's expenditure share in firm revenue.

Following Peters and Taylor (2017), I measure R&D investment as total R&D expenditure (XRD) and firm-specific intangible investment as 30% of Selling, General, and Administrative expenses net of R&D (XSGA − XRD). SG&A encompasses a broad range of expenditures including advertising, employee compensation, and general operational costs. Once R&D is netted out, the remaining SG&A primarily reflects advertising and employee-related costs, most of which are routine operating expenses consumed in the current period. A portion, however, accumulates within the firm as lasting intangible capital: advertising builds brand recognition, and employee compensation develops organizational knowledge and managerial talent. The 30% fraction captures this accumulating component—firm-specific intangible capital composed of brand value and organizational capital—with the remainder treated as routine operating costs that generate no lasting intangible value.[7]

3. Dynamic Response to Innovation Shocks

This section asks whether the synergy–friction logic that organises the static cross-section also governs how firms absorb innovation shocks dynamically. Related scope furnishes synergistic deployment outlets: a technological gain originating in one product-market cluster can be redeployed across neighbouring clusters within the same operational domain, so firms with broader related scope possess a larger appropriation channel for any

[6]Table A5 reports elasticity estimates including alternative estimators Ackerberg *et al.* (2015) and Gandhi *et al.* (2020).

[7]The two components of firm-specific intangibles—brand value and organizational capital—could in principle be identified separately, since advertising expenditure is reported as a distinct line item (XAD) in Compustat. However, XAD is sparsely reported, and conditioning on non-missing values would disproportionately retain firms with large advertising budgets, introducing a systematic selection toward advertising-intensive firms.

given innovation. Distant scope generates the opposite force: operating across operationally separable business segments fragments managerial attention, dilutes the marginal incentive to concentrate innovative effort within any single segment, and accumulates firm-specific intangibles—brand reputation and organisational capital—that insulate the firm from the competitive consequences, and therefore the appropriable rents, of innovation. Two predictions follow. Firms with low distant scope (single-segment, focused operations) should respond more aggressively than firms with high distant scope (multi-segment, diversified operations). Firms with broad related scope should respond more aggressively than firms with narrow related scope. I test these predictions using local projections (Jordà, 2005) with the patent-value innovation shocks of Kogan *et al.* (2017). The patent-value measure isolates the unexpected component of grant-related news within a short event window around grant dates, and grant timing is largely predetermined relative to the firm's contemporaneous strategic decisions. The local-projection specification is

$$\Delta_h Y_{ijt} = \alpha_h + \beta^h_{\text{High}}\, V_{it}\, D_{it} + \beta^h_{\text{Low}}\, V_{it}\,(1 - D_{it}) + \boldsymbol{\Gamma}_h^\top\, \mathbf{X}_{i,t-1} + \delta_h\, Y_{i,t-1} + \theta_j + \lambda_t + u_{ijt}, \quad (4)$$

where $\Delta_h Y_{ijt} \equiv Y_{i,t+h} - Y_{it}$ is the h-period log change in outcome Y for firm i in industry j, $V_{it} = \sum_{p \in \mathcal{P}_{it}} \text{PatentValue}_{p,t}$ is the firm-year sum of patent values, and $D_{it} \in \{0, 1\}$ is an indicator for the high-scope group along one of the two scope dimensions. The shock enters as $\log(1 + V_{it})$ to retain firm-years with zero patent grants. I estimate the specification separately for each scope dimension. For the distant-scope split, $D_{it} = 1$ identifies multi-segment firms (high distant scope) and $D_{it} = 0$ identifies single-segment, focused firms (low distant scope). For the related-scope split, $D_{it} = 1$ identifies firms whose `d2vscope` lies at or above the sample median (broad related scope) and $D_{it} = 0$ identifies firms below the median (narrow related scope). The coefficients β^h_{High} and β^h_{Low} trace the impulse responses of the two groups to a one-unit increase in log patent value at horizon h. The vector $\mathbf{X}_{i,t-1}$ collects lagged firm-level controls, $Y_{i,t-1}$ absorbs pre-shock outcome levels, and θ_j and λ_t denote industry and year fixed effects. Scope status is fixed at each firm's pre-shock classification and held constant throughout the estimation window.

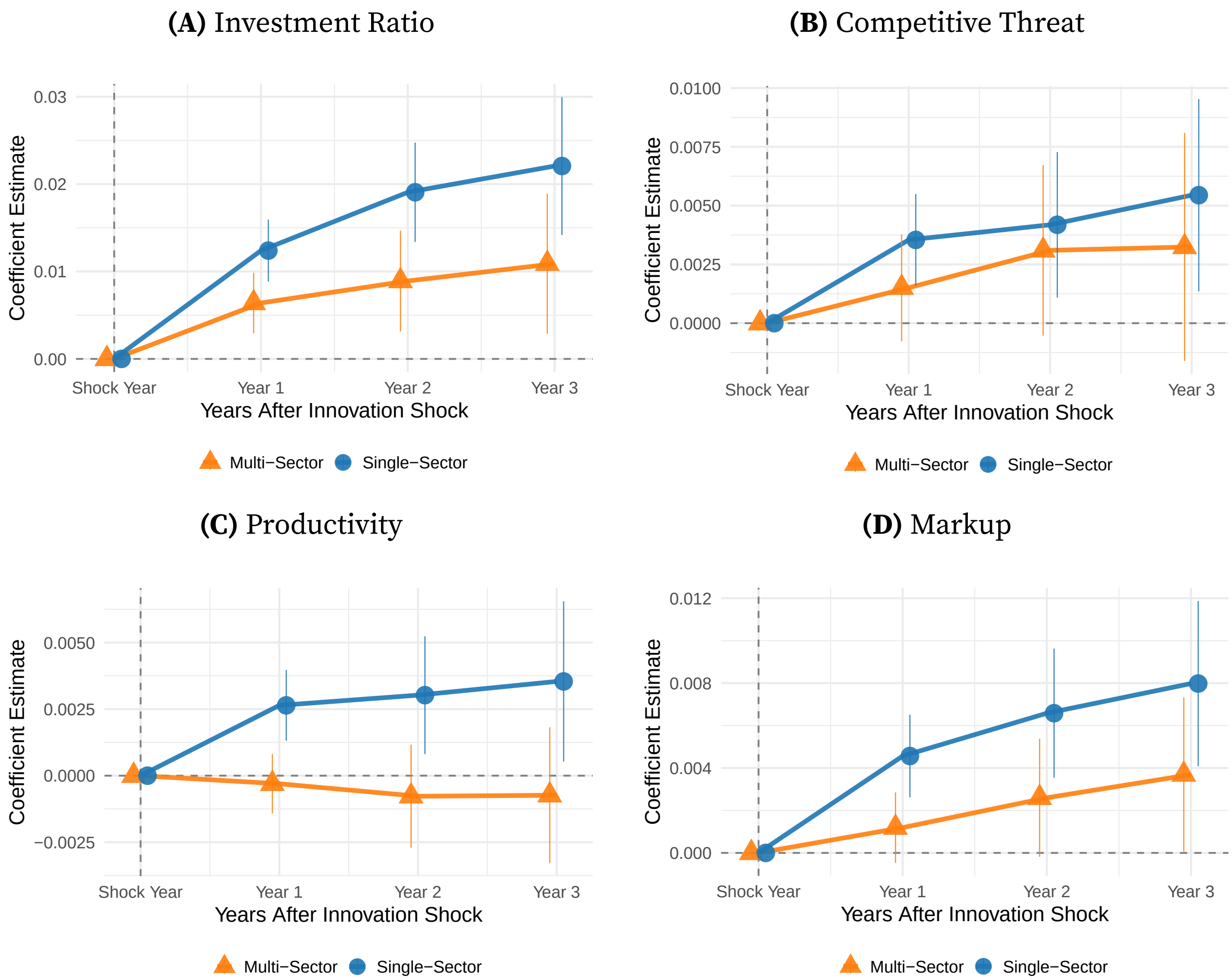

Figure 3. Dynamic Responses to Innovation Shocks by Distant Scope (Compustat Business Segments)

Note: The figure plots local-projection impulse responses to a patent-value innovation shock, estimated by equation (4), separately for two groups defined by distant scope. *Single-Sector* firms (low distant scope, focused) report one Compustat business segment in the year prior to the shock; *Multi-Sector* firms (high distant scope, diversified) report two or more. The shock enters as $\log(1 + V_{it})$, where V_{it} is the firm-year sum of Kogan *et al.* (2017) patent values, to retain firm-years with zero patent grants. Outcomes are defined as the h-year log change, $\Delta_h \log(Y)$, and all controls enter in logs; productivity and markup use the Olley–Pakes baseline. The sample covers fiscal years 1990–2019 and excludes utilities, finance, and firm-years with missing or non-positive R&D, SG&A, employment, or sales. Scope status is fixed at the pre-shock classification and held constant over the projection horizon. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry–year level.

Figure 3 contrasts the impulse responses of single-segment firms (low distant scope, focused) and multi-segment firms (high distant scope, diversified). The two groups separate on every margin, and the separation widens monotonically over the three-year horizon. The investment ratio rises sharply for single-segment innovators, indicating a shift in intangible-investment composition toward R&D, while multi-segment innovators exhibit a much weaker reallocation; the single-segment response is roughly two and a half times the multi-segment response at the three-year horizon (Panel A). Single-segment firms attract significant rival entry into their product space, while multi-segment firms show no measurable change in competitive pressure (Panel B). Productivity rises steadily for single-segment firms but remains flat—if anything slightly negative—for multi-segment firms, so the productivity gap between the two groups widens persistently (Panel C). The markup response follows the same pattern: single-segment firms command appreciably higher price–cost margins after the shock, while multi-segment firms exhibit a positive but materially smaller response (Panel D).

Figure 4 reports the parallel exercise for related scope, contrasting firms above and below the sample median of `d2vscope`. The separation between the two groups is sharper than under the distant-scope split. Broad-related firms shift their intangible-investment composition aggressively toward R&D, with the response approximately twice that of narrow-related firms at the three-year horizon (Panel A). Competitive pressure rises significantly for broad-related innovators but declines for narrow-related innovators, suggesting that rivals retreat from the product space of firms whose related scope is too thin to sustain competitive engagement (Panel B). Productivity gains are positive and rising for broad-related firms but turn persistently negative for narrow-related firms over the three-year horizon (Panel C), indicating that narrow-related innovators incur the input costs of R&D without realising the scale benefits that a broader portfolio of synergistic sub-markets would deliver. Markups rise robustly for broad-related firms and remain near zero, with a slight negative tilt, for narrow-related firms (Panel D).

The two splits deliver a coherent dynamic picture organised by the synergy–friction framework. Firms whose scope structure favours synergy—low distant scope or broad related scope—absorb innovation shocks aggressively, shifting intangible-investment composition toward R&D, attracting rival entry, and converting technological gains into persistent productivity and markup advantages. Firms whose structure is dominated by distant scope, or whose related scope is too narrow to sustain a synergistic portfolio, exhibit muted responses on every margin, with productivity gains attenuated or reversed and markup gains compressed.[8]

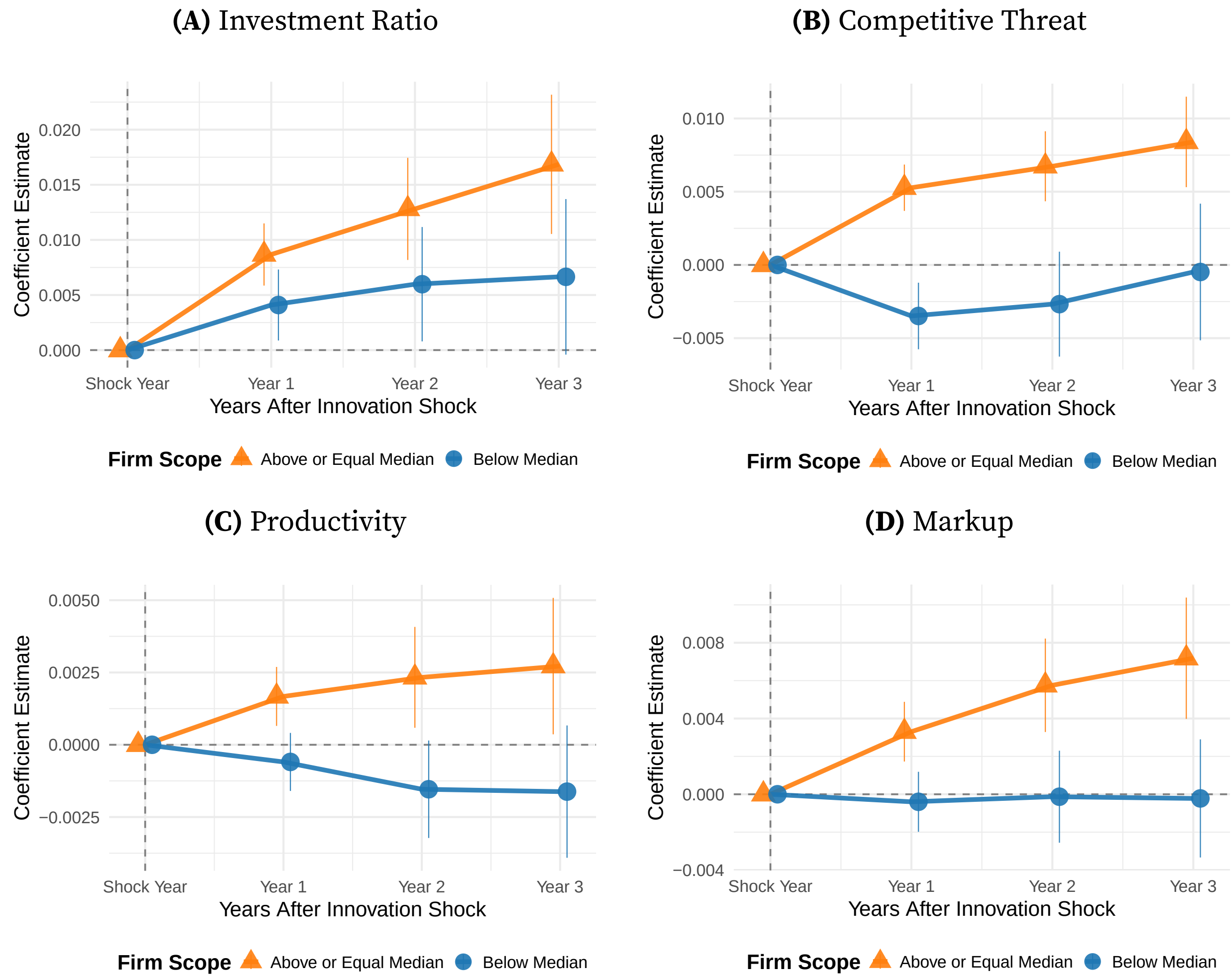

Figure 4. Dynamic Responses to Innovation Shocks by Related Scope (Hoberg–Phillips Product-Market Clusters)

Note: The figure plots local-projection impulse responses to a patent-value innovation shock, estimated by equation (4), separately for two groups defined by related scope. *Above or Equal Median* firms (broad related scope) are firm-years whose Hoberg–Phillips `d2vscope` lies at or above the sample median in the year prior to the shock; *Below Median* firms (narrow related scope) lie below the median. The shock enters as $\log(1 + V_{it})$, where V_{it} is the firm-year sum of Kogan *et al.* (2017) patent values, to retain firm-years with zero patent grants. Outcomes are defined as the h-year log change, $\Delta_h \log(Y)$, and all controls enter in logs; productivity and markup use the Olley–Pakes baseline. The sample covers fiscal years 1990–2019 and excludes utilities, finance, and firm-years with missing or non-positive R&D, SG&A, employment, or sales. Scope status is fixed at the pre-shock classification and held constant over the projection horizon. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry–year level.

[8] Appendix Figure A1 and Figure A2 replicate the analysis using productivity and markup estimates based on Ackerberg *et al.* (2015) and Gandhi *et al.* (2020), and the qualitative patterns are unchanged.

4. Conclusion

This paper shows that the relationship between firm scope and firm performance is not a scalar fact about diversification but a structural one about how scope is composed. Distinguishing the operationally distant scope captured by the Compustat segment count from the closely related product-market scope recovered by the Hoberg–Phillips text-based measure isolates two distinct economic forces operating on the same firm. Broad related scope generates synergy: firms with broader related portfolios command higher markups, run higher productivity, allocate a larger share of intangible investment to R&D, and absorb innovation shocks aggressively on every margin examined. High distant scope generates friction: firms diversified across operationally separable business segments command lower markups, run lower productivity, tilt their intangible composition away from R&D and toward firm-specific brand and organizational capital, and convert a given innovation shock into materially weaker gains.

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Appendices

A. Dataset

Table A1. Firm Scope: Compustat vs. Hoberg-Phillips

Company	**Scope (Compustat)**	**Scope (Hoberg-Phillips)**
TESLA INC	Energy Generation & Storage Automotive	Batteries Automotive (Brakes, Trim, Axle, Engines, Chassis) Automotive Safety (Airbags) Car Dealerships Energy / Cogeneration Utilities / Electric Power Smart Metering / Grid Tech Power Electronics / Voltage Solar / Renewable Energy Hardware & Software Solutions Ticketing / Scanning (Software/Systems)

B. Measurement Details

B.1. Production Function Estimation

The sample is restricted to observations satisfying `sale` > 0, `ppent` > 0, `cogs` > 0, `emp` > 0, `capx` > 0, and `cogs` $<$ `sale`, where the final restriction ensures positive gross profits. All nominal variables are deflated to real terms prior to estimation: output by the GDP deflator, intermediate inputs by the Producer Price Index, and capital stock and capital expenditure by the Gross Private Domestic Investment deflator.[9]

Robustness: Ackerberg *et al.* (2015). Ackerberg *et al.* (2015) identify the first stage estimates only the composite function $\phi_t(l_{it}, k_{it}, m_{it}, i_{it})$ nonparametrically, and β_l, β_k, and

[9] GDP deflator: https://fred.stlouisfed.org/series/A191RD3A086NBEA. PPI: https://fred.stlouisfed.org/series/WPUID61. Investment deflator: https://fred.stlouisfed.org/series/A006RD3A086NBEA.

β_m are identified jointly in the second stage via moment conditions on the productivity innovation, using lagged values of variable inputs as instruments to exploit the timing assumption that these inputs are determined before ω_{it} realizes.

Robustness: Gandhi *et al.* (2020). Gandhi *et al.* (2020) identify the output elasticity of the flexible input from the firm's static cost-minimization condition, without imposing functional form restrictions on the production function. Under perfect competition in intermediate input markets, cost minimization implies that the output elasticity of intermediates equals the revenue share of intermediate input expenditure, after correcting for measurement error in observed output:

$$\frac{\partial \ln Q_{it}}{\partial \ln M_{it}} = \frac{P_{it}^{M} M_{it}}{P_{it} Q_{it}} \cdot \mathcal{E}(\epsilon_{it}), \tag{5}$$

where $\mathcal{E}(\epsilon_{it}) \equiv \mathbb{E}[e^{\epsilon_{it}}]$ is the measurement-error correction. This share equation identifies the flexible-input elasticity in the first stage without functional-form restrictions on the production function. The remaining input coefficients are recovered in the second stage using moment conditions on the productivity innovation, analogous to the OP and ACF second stages.

C. Additional Figures and Empirical Results

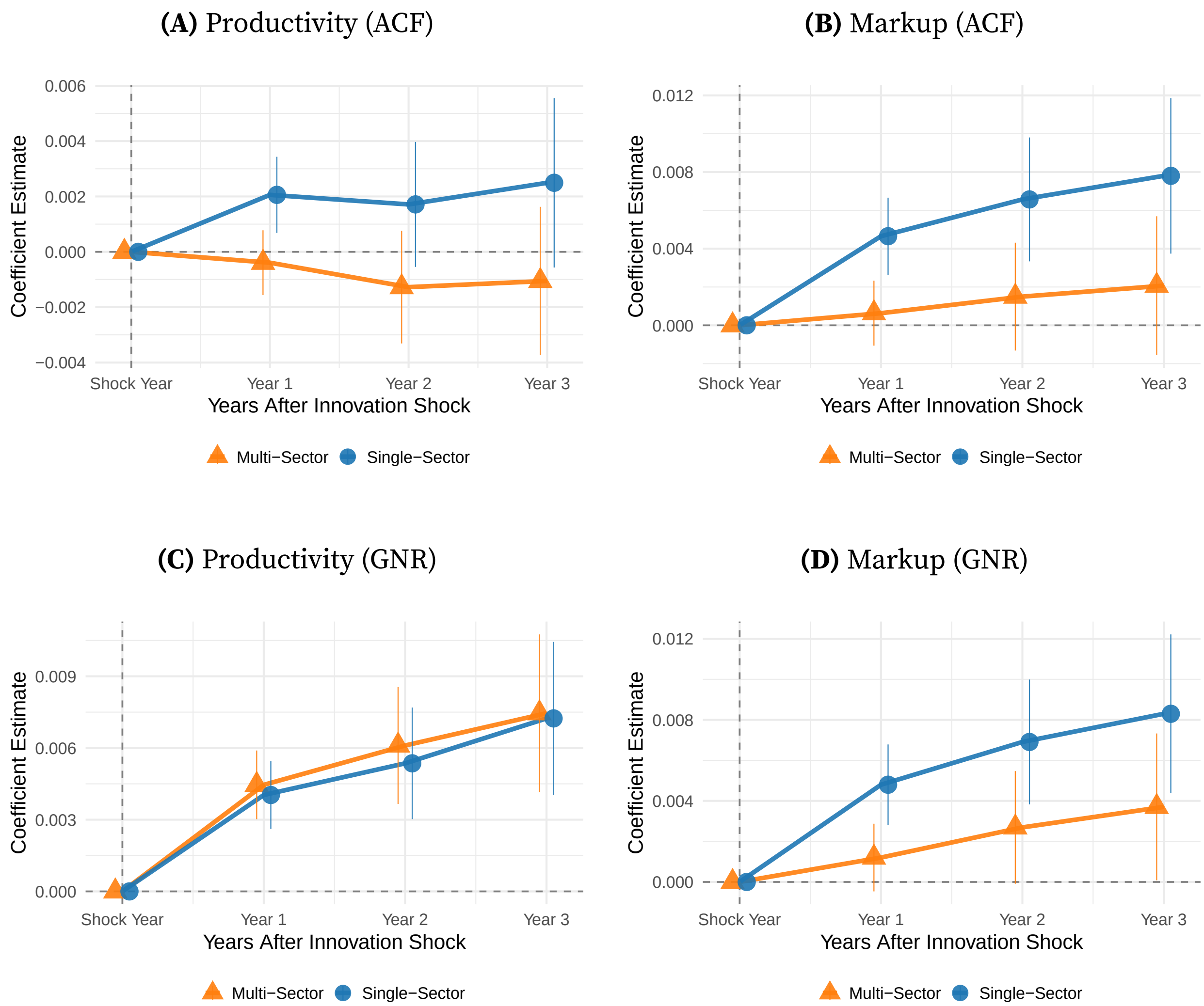

Figure A1. Dynamic Responses to Innovation Shocks by Distant Scope (Compustat Business Segments): ACF and GNR

Note: Panels (A) and (B) report estimates based on Ackerberg *et al.* (2015); Panels (C) and (D) report estimates based on Gandhi *et al.* (2020). The figure plots local-projection impulse responses to a patent-value innovation shock, estimated by equation (4), separately for two groups defined by distant scope. *Single-Sector* firms (low distant scope, focused) report one Compustat business segment in the year prior to the shock; *Multi-Sector* firms (high distant scope, diversified) report two or more. The shock enters as $\log(1+V_{it})$, where V_{it} is the firm-year sum of Kogan *et al.* (2017) patent values, to retain firm-years with zero patent grants. Outcomes are defined as the h-year log change, $\Delta_h \log(Y)$, and all controls enter in logs. The sample covers fiscal years 1990–2019 and excludes utilities, finance, and firm-years with missing or non-positive R&D, SG&A, employment, or sales. Scope status is fixed at the pre-shock classification and held constant over the projection horizon. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry–year level.

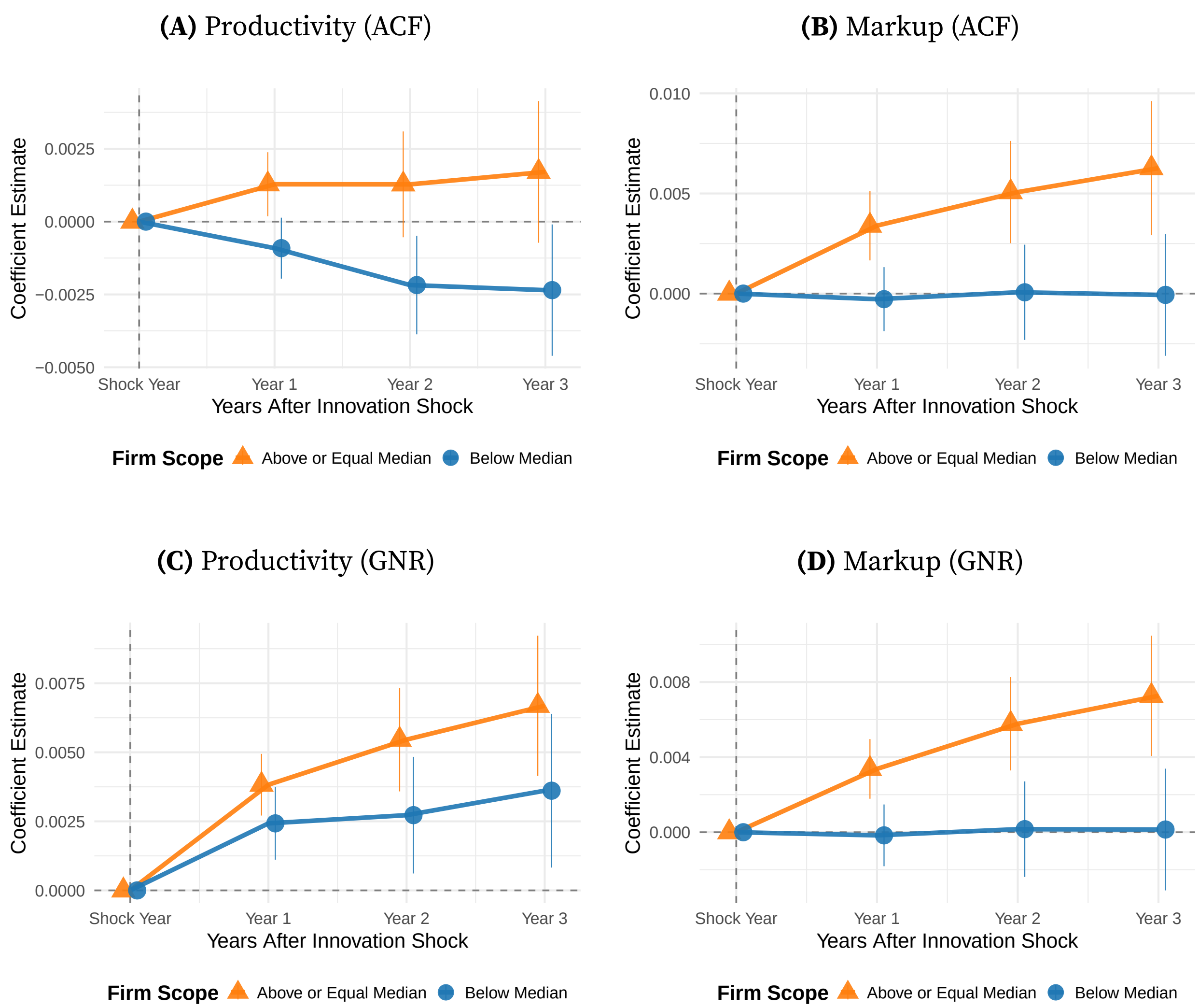

Figure A2. Dynamic Responses to Innovation Shocks by Related Scope (Hoberg–Phillips Product-Market Clusters): ACF and GNR

Note:: Panels (A) and (B) report estimates based on Ackerberg *et al.* (2015); Panels (C) and (D) report estimates based on Gandhi *et al.* (2020). The figure plots local-projection impulse responses to a patent-value innovation shock, estimated by equation (4), separately for two groups defined by related scope. *Above or Equal Median* firms (broad related scope) are firm-years whose Hoberg–Phillips `d2vscope` lies at or above the sample median in the year prior to the shock; *Below Median* firms (narrow related scope) lie below the median. The shock enters as $\log(1 + V_{it})$, where V_{it} is the firm-year sum of Kogan *et al.* (2017) patent values, to retain firm-years with zero patent grants. Outcomes are defined as the h-year log change, $\Delta_h \log(Y)$, and all controls enter in logs. The sample covers fiscal years 1990–2019 and excludes utilities, finance, and firm-years with missing or non-positive R&D, SG&A, employment, or sales. Scope status is fixed at the pre-shock classification and held constant over the projection horizon. Vertical bars denote 95% confidence intervals based on standard errors clustered at the industry–year level.

Table A2. Regression Results: Compustat

Panel A: *Markup* and *Productivity*

	Pooled OLS		Two-way FE	
	(2) *Markup*	(3) *Productivity*	(6) *Markup*	(7) *Productivity*
Segment	-0.038^{***}	-0.056^{***}	-0.041^{***}	-0.044^{***}
	(0.004)	(0.008)	(0.004)	(0.003)
Num. Obs.	40,008	40,008	40,008	40,008
Adj. R^2	0.268	0.197	0.279	0.833
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Panel B: *Investment Ratio* and *Fluidity*

	Pooled OLS		Two-way FE	
	(1) *Transferable/Embedded*	(4) *Comp. Threat*	(5) *Transferable/Embedded*	(8) *Comp. Threat*
Segment	-0.011^{***}	-0.037^{***}	-0.014^{***}	-0.017^{**}
	(0.002)	(0.006)	(0.002)	(0.006)
Num. Obs.	42,831	31,318	42,831	31,318
Adj. R^2	0.977	0.208	0.978	0.305
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Notes: Each column reports coefficients from a separate regression. Standard errors are clustered by *firm id* (gvkey) in parentheses. Significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Columns 1–4: Pooled OLS specifications; Columns 5–8: Two-way fixed effects (year and industry). Covariates include sale, xrd, and xsga.

Table A3. Regression Results: HP Scope

Panel A: *Markup* and *Productivity*

	Pooled OLS		Two-way FE	
	(2) *Markup*	(3) *Productivity*	(6) *Markup*	(7) *Productivity*
HP-Scope	0.010^{***}	0.023^{***}	0.008^{***}	0.012^{***}
	(0.001)	(0.002)	(0.001)	(0.001)
Num. Obs.	54,046	54,046	54,046	54,046
Adj. R^2	0.038	0.051	0.071	0.777
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Panel B: *Investment Ratio* and *Fluidity*

	Pooled OLS		Two-way FE	
	(1) *Transferable/Embedded*	(4) *Comp. Threat*	(5) *Transferable/Embedded*	(8) *Comp. Threat*
HP-Scope	0.039^{***}	0.202^{***}	0.037^{***}	0.248^{***}
	(0.002)	(0.006)	(0.002)	(0.006)
Num. Obs.	58,279	51,975	58,279	51,975
Adj. R^2	0.077	0.161	0.153	0.340
Covariates	Yes	Yes	Yes	Yes
FE: *Year*	No	No	Yes	Yes
FE: *Industry*	No	No	Yes	Yes

Notes: Each column reports coefficients from a separate regression. Standard errors are clustered by *firm id* (gvkey) in parentheses. Significance levels: $^{+}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Covariates include sale, at, xrd, and xsga.

Table A4. Summary Statistics

Panel A: Compustat–Segment Merged

Variable	N	Mean	SD	P25	Median	P75
Sale	43,007	3,597,348	16,753,747	29,041	166,258	1,120,375
Emp	43,007	10,466	37,694	154	708	4,482
AT	43,007	5,101,380	26,098,750	37,743	211,989	1,398,987
XRD	43,007	180,872	867,769	2,154	10,697	49,903
XSGA	43,007	721,384	2,891,804	14,998	59,082	266,119
Productivity (OLS)	40,008	21.055	24.665	8.054	11.264	25.035
Markup (OLS)	39,603	1.549	1.056	0.986	1.205	1.674
Productivity (ACF)	31,937	31.702	53.825	6.469	8.471	36.590
Markup (ACF)	28,218	1.287	0.785	0.889	1.082	1.423
Productivity (GNR)	40,404	11.991	23.298	0.891	1.549	7.952
Markup (GNR)	39,680	1.300	0.935	0.825	1.011	1.388
Competitive Threat	31,318	6.476	3.120	4.203	6.020	8.183

Panel B: Hoberg–Phillips Scope Merged

Variable	N	Mean	SD	P25	Median	P75
Sale	35,964	2,171,465	10,468,548	25,104	135,367	826,899
Emp	35,964	6,440	23,946	133	571	3,212
AT	35,964	3,083,646	19,749,630	30,984	167,030	977,718
XRD	35,964	127,011	736,443	1,813	8,868	39,625
XSGA	35,964	476,449	2,256,258	13,128	50,168	208,927
Productivity (OLS)	33,376	20.556	24.630	7.890	11.082	23.682
Markup (OLS)	33,035	1.545	1.048	0.986	1.207	1.673
Productivity (ACF)	26,866	31.199	53.967	6.398	8.439	34.253
Markup (ACF)	23,712	1.295	0.795	0.895	1.089	1.434
Productivity (GNR)	33,706	11.672	23.026	0.883	1.473	7.212
Markup (GNR)	33,128	1.298	0.928	0.826	1.014	1.391
Competitive Threat	30,097	6.451	3.104	4.189	6.005	8.156

Notes: This table reports summary statistics for the three estimation samples. Panel A covers Compustat Fundamentals Annual restricted to firm-years with positive sales, employment, R&D (XRD), and SG&A (XSGA). Panel B merges Panel A with the Compustat Segment database, retaining all countries. Panel C restricts Panel B to U.S.-incorporated firms (*fic* = USA).

Table A5. Sector-Level Elasticities: OP, ACF, and GNR

		OP				*ACF*				*GNR*			
NAICS	Sector	β_M	β_K	β_L	RTS	β_M	β_K	β_L	RTS	β_M	β_K	β_L	RTS
11	Agriculture, Forestry & Fishing	0.834	0.032	0.100	0.966	0.612	0.146	0.137	0.894	0.676	0.209	0.148	1.030
21	Mining & Oil and Gas Extraction	0.719	0.118	0.050	0.887	0.692	0.280	0.001	0.972	0.500	0.386	0.111	0.997
23	Construction	0.931	0.023	0.012	0.966	—	—	—	—	0.826	0.018	0.131	0.975
31	Food & Beverage Manufacturing	0.853	0.029	0.066	0.947	—	—	—	—	0.659	0.148	0.189	0.996
32	Chemical & Paper Manufacturing	0.612	0.019	0.275	0.907	0.351	0.094	0.501	0.946	0.489	0.106	0.399	0.994
33	Machinery & Computer Manufacturing	0.724	0.027	0.184	0.935	0.692	0.076	0.199	0.967	0.617	0.091	0.315	1.020
42	Wholesale Trade	0.832	0.046	0.086	0.963	0.504	0.065	0.330	0.900	0.790	0.054	0.160	1.000
44	Retail Trade I	0.782	0.043	0.106	0.931	—	—	—	—	0.728	0.120	0.203	1.050
45	Retail Trade II	0.780	0.046	0.064	0.889	0.466	0.094	0.290	0.850	0.681	0.154	0.207	1.040
48	Transportation & Warehousing	—	—	—	—	0.628	0.120	0.113	0.860	0.755	0.196	0.065	1.020
51	Information	0.490	0.059	0.336	0.884	—	—	—	—	0.354	0.128	0.468	0.951
53	Real Estate & Rental	—	—	—	—	—	—	—	—	0.363	0.188	0.306	0.857
54	Professional & Technical Services	0.634	0.054	0.183	0.871	0.119	0.175	0.583	0.877	0.580	0.132	0.310	1.020
56	Administrative & Support Services	0.766	0.053	0.078	0.897	—	—	—	—	0.655	0.170	0.185	1.010
61	Educational Services	0.708	0.109	0.043	0.860	0.446	0.137	0.281	0.864	0.486	0.211	0.287	0.984
62	Health Care & Social Assistance	0.819	0.052	0.011	0.882	0.396	0.122	0.317	0.834	0.708	0.110	0.117	0.935
71	Arts, Entertainment & Recreation	0.871	0.054	0.023	0.948	0.424	0.246	0.177	0.847	0.649	0.179	0.135	0.963
72	Accommodation & Food Services	0.819	0.095	0.021	0.935	0.548	0.201	0.185	0.934	0.739	0.197	0.110	1.040
81	Other Services	—	—	—	—	0.796	0.053	0.105	0.954	0.665	0.102	0.139	0.905
99	Nonclassifiable Establishments	0.748	0.007	0.159	0.915	0.418	0.084	0.393	0.894	0.617	0.074	0.251	0.942

Notes: β_M, β_K, and β_L denote elasticities with respect to materials (flexible input), capital, and labor (both dynamic inputs), respectively. RTS $\equiv \beta_M + \beta_K + \beta_L$ is the implied returns to scale. OP refers to olleypakes1996; ACF refers to ackerberg2015identification; GNR refers to gandhi2020identification. Missing values (—) indicate that the estimation did not produce valid positive elasticities or RTS $\notin [0.8, 1.2]$.